

Super Store Sales Dashboard

A Power BI Sales Performance Analysis Project

1. Project Overview

This project analyzes transactional sales data from a retail superstore to uncover trends in revenue, profitability, and order volume across product categories, customer segments, shipping methods, payment methods, and geographic regions. The goal was to build an interactive Power BI dashboard that helps stakeholders quickly identify top-performing categories, seasonal sales patterns, and areas where profitability lags behind sales volume, in order to guide merchandising, shipping, and regional strategy decisions.

2. Dataset Summary

The dataset contains order-level retail transactions with the following key fields:

- **Order details:** Order Date, Region, State
- **Product details:** Category, Sub-Category (e.g. Chairs, Binders, Phones)
- **Customer details:** Segment (Consumer, Corporate, Home Office)
- **Fulfillment details:** Ship Mode (Standard, Second, First, Same Day Class), Payment Mode (COD, Online, Card)
- **Performance metrics:** Sales, Profit, Quantity

3. Data Preparation using Power Query

Before analysis, the raw data was cleaned and shaped inside Power BI's Power Query Editor:

- **Data Import:** Loaded the raw sales transactions into Power BI Desktop.
- **Data Cleaning:** Checked for and handled missing or inconsistent values across key columns.
- **Column Standardization:** Reviewed and standardized category, sub-category, region, and payment mode labels for consistent grouping.
- **Date Handling:** Built out a date hierarchy (Year / Month) from Order Date to support the Monthly Sales and Profit by YoY visuals.
- **Data Modeling:** Structured the tables and relationships needed to support cross-filtering between the Region slicer and all visuals on the dashboard.

4. Data Analysis using DAX

Core measures (Sales, Profit, Quantity) were built with DAX to power the dashboard's KPI cards and charts, and were then broken down along several dimensions to answer key business questions:

- **Sales & Profit by Category / Sub-Category** – identifying which product lines drive revenue vs. which drive profit.

- **Monthly Sales by YoY and Monthly Profit by YoY** – time-intelligence measures comparing performance across years to surface seasonality.
- **Sales by Segment, Payment Mode, and Region** – donut-chart breakdowns of the customer and transaction mix.
- **Sales by Ship Mode** – comparing order volume across Standard, Second, First, and Same Day shipping classes.
- **Sum of Sales and Sum of Profit by State** – a geographic map measure for regional performance.

5. Dashboard in Power BI

The dashboard brings these measures together into a single interactive view, with a Region slicer at the top that cross-filters every visual — KPI cards for Sales, Quantity, and Profit; a Monthly Sales and Monthly Profit trend by year; donut breakdowns by Segment, Payment Mode, and Region; a Sales by Ship Mode comparison; Sales by Category and Sub-Category rankings; and a state-level map of Sales and Profit.

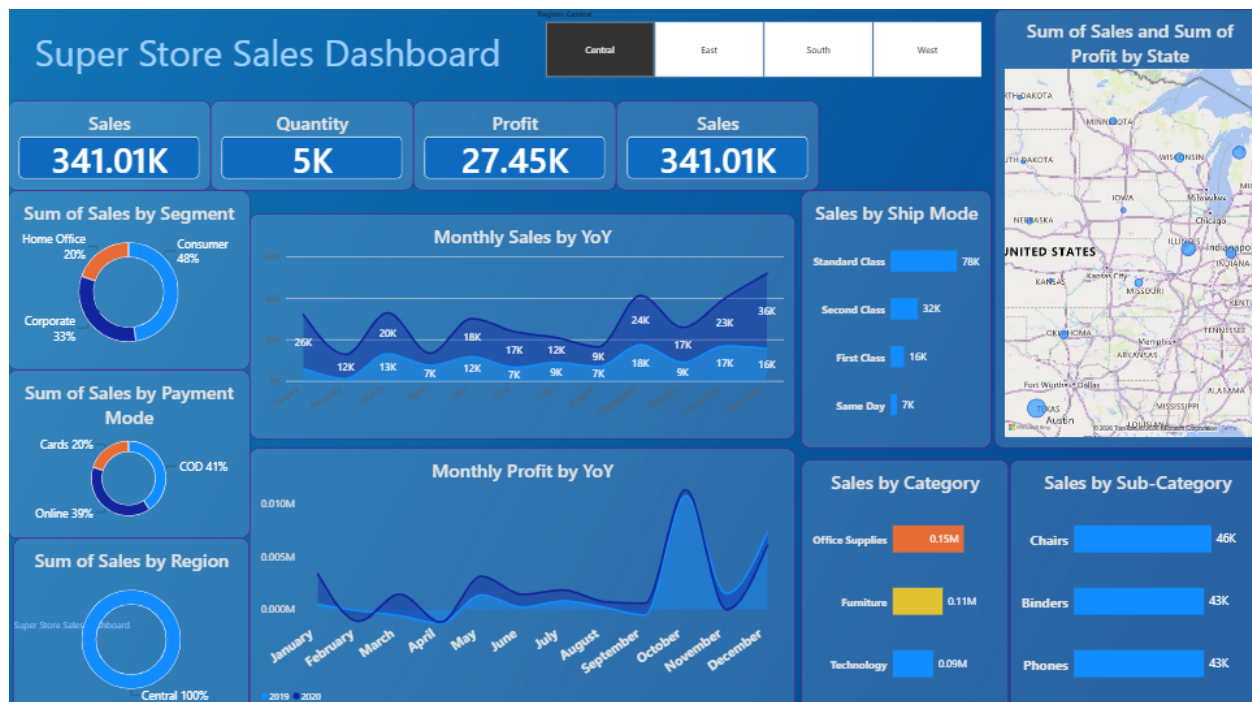


Figure: Super Store Sales Dashboard — Overview page (Power BI), filtered to the Central region.

6. Business Recommendations

- **Grow the Home Office segment** – it trails Consumer (48%) and Corporate (33%) at just 20% of sales, and represents room for targeted campaigns.
- **Encourage digital payments** – COD still accounts for 41% of transactions vs. 39% Online and 20% Card; shifting more volume online can reduce collection overhead.
- **Revisit Technology profitability** – Technology contributes less profit (0.09M) than Office Supplies (0.15M) despite typically higher price points, suggesting discounting or cost pressures worth investigating.

- **Promote express shipping selectively** – Standard Class dominates fulfillment (78K orders) while Same Day remains niche (7K); a targeted upsell to faster shipping tiers could lift margins.
- **Plan for the Q4 peak** – both sales and profit trend upward into October–December, so inventory and staffing plans should anticipate this seasonal surge.
- **Double down on top sub-categories** – Chairs, Binders, and Phones lead sub-category sales and are strong candidates for featured placement or bundling.